

DAMON (nano) Conference Opening

SJ Park <sj@kernel.org> <sjpark@crusoe.ai>



LINUX
PLUMBERS CONFERENCE

PRAGUE, CZECHIA / OCT. 5-7, 2026



DAMON (nano) Conference

- First DAMON conference
- 45 Minutes in total
 - Extended to hallway and evening events
- Five talks, eight authors
- (Micro) conference next year, hopefully

DAMON in Nutshell

DAMON in One Sentence

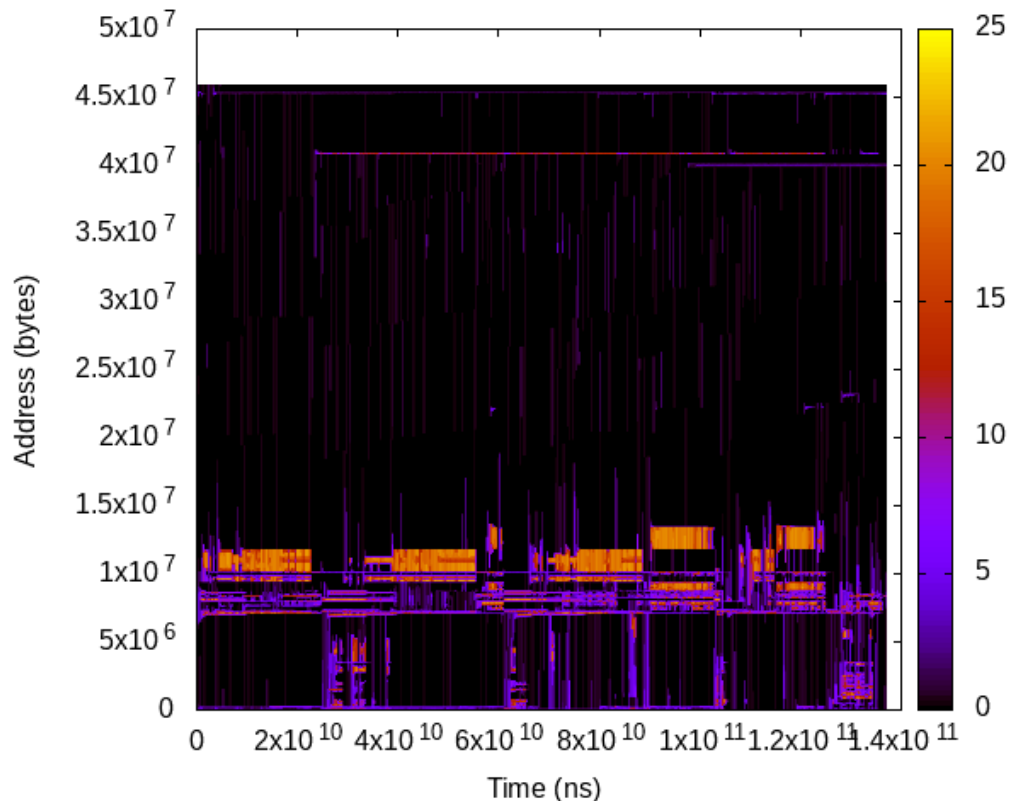
“DAMON is a Linux kernel subsystem for efficient access monitoring and access-aware system operations.”

High Level Implementation

- Spawn a kernel thread that does
- See if each page is accessed, every 5ms
- Inform users the findings, every 100ms
- Time intervals can be [auto-]tuned

Providing Access Information

- Location
- Frequency
- Stability
- Recency



Access pattern heatmap of
Splash2x/Raytrace

Design

- Goals: Accurate, Lightweight, Tunable and Automated
- Mechanisms: Sampling, Adaptive regions, Sysfs interface, Feedback loops
- Results
 - $<0.1\%$ CPU overhead on products

Availability and Usages

- Available on ≥ 5.15 upstream kernels
- Enabled on most [distro](#) kernels
- Being used for products and researches

DAMOS: Second Face of DAMON

- DAMon-based Operation Schemes
 - “Page out cold memory”
 - “Migrate hot/cold pages to DRAM/CXL node”
- Turn DAMON into access-aware memory management system operations engine

Community

- Public channels
 - Mailing list: damon@lists.linux.dev
 - Project website: <https://damonitor.github.io/>
- Private channels
 - Maintainer email: sj@kernel.org
 - DAMON Beer/Coffee/Tea [Meetup](#)

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